

solution 3 (resubmission)

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BS24B006

Q.1

on uniprot site

UniProt Tools UniProtKB Advanced List Search

P21796 · VDAC1_HUMAN

Protein ⁱ	Non-selective voltage-gated ion channel VDAC1	Amino acids	283 (go to sequence)
Gene ⁱ	VDAC1	Protein existence ⁱ	Evidence at protein level
Status ⁱ	UniProtKB reviewed (Swiss-Prot)	Annotation score ⁱ	5/5
Organism ⁱ	Homo sapiens (Human)		

Tools Download Add Add a publication Entry feedback

Entry Variant viewer Feature viewer Genomic coordinates Publications External links

Functionⁱ

Non-selective voltage-gated ion channel that mediates the transport of anions and cations through the mitochondrion outer membrane and plasma membrane (PubMed:10661876, PubMed:11845315, PubMed:18755977, PubMed:30061676, PubMed:8420959). The channel at the outer mitochondrial membrane allows diffusion of small hydrophilic molecules; in the plasma membrane it is involved in cell volume regulation and apoptosis (PubMed:10661876, PubMed:11845315, PubMed:18755977, PubMed:8420959). It adopts an open conformation at low or zero membrane potential and a closed conformation at potentials above 30-40 mV (PubMed:10661876, PubMed:18755977, PubMed:8420959). The open state has a weak anion selectivity whereas the closed state is cation-selective.

sequence:

<https://rest.uniprot.org/uniprotkb/P21796.txt>

at the very end theres a sequence line:

Plaintext

```
SQ   SEQUENCE   283 AA;  30773 MW;
89BA3378B04020D5 CRC64;
MAVPPTYADL GKSARDVFTK GYGFGLIKLD LTKKSENGLE
FTSSGSANTE TTKVTGSLET
KYRWTEYGLT FTEKWNTDNT LGTEITVEDQ LARGLKLTFD
SSFSPNTGKK NAKIKTGYKR
```

EHINLGCDMD FDIAGPSIRG ALVLGYEGWL AGYQMNFETA

KSRVTQSNFA VGYKTDEFQL

HTNVNDGTEF GGSIIYQKVNK KLETAVNLAW TAGNSNTRFG

IAAKYQIDPD ACFSAKVNNS

SLIGLGYTQT LKPGIKLTLS ALLDGKNVNA GGHKLGLGLE

FQA

Function¹

Non-selective voltage-gated ion channel that mediates the transport of anions and cations through the mitochondrion outer membrane and plasma membrane (PubMed:[10661876](#), PubMed:[11845315](#), PubMed:[18755977](#), PubMed:[30061676](#), PubMed:[8420959](#)).

The channel at the outer mitochondrial membrane allows diffusion of small hydrophilic molecules; in the plasma membrane it is involved in cell volume regulation and apoptosis (PubMed:[10661876](#), PubMed:[11845315](#), PubMed:[18755977](#), PubMed:[8420959](#)).

It adopts an open conformation at low or zero membrane potential and a closed conformation at potentials above 30-40 mV (PubMed:[10661876](#), PubMed:[18755977](#), PubMed:[8420959](#)).

The open state has a weak anion selectivity whereas the closed state is cation-selective (PubMed:[18755977](#), PubMed:[8420959](#)).

Binds various signaling molecules, including the sphingolipid ceramide, the phospholipid phosphatidylcholine, and the sterols cholesterol and oxysterol (PubMed:[18755977](#), PubMed:[31015432](#)).

In depolarized mitochondria, acts downstream of PRKN and PINK1 to promote mitophagy or prevent apoptosis; polyubiquitination by PRKN promotes mitophagy, while monoubiquitination by PRKN decreases mitochondrial calcium influx which ultimately inhibits apoptosis (PubMed:[32047033](#)).

May participate in the formation of the permeability transition pore complex (PTPC) responsible for the release of mitochondrial products that triggers apoptosis (PubMed:[15033708](#), PubMed:[25296756](#)).

May mediate ATP export from cells (PubMed:[30061676](#)).

Part of a complex composed of HSPA9, ITPR1 and VDAC1 that regulates mitochondrial calcium-dependent apoptosis by facilitating calcium transport from the ER lumen to the mitochondria intermembrane space thus providing calcium for the downstream calcium channel MCU that directly releases it into mitochondria matrix (By similarity).

Mediates cytochrome c efflux (PubMed:[20230784](#)).  By Similarity  10 Publications

Catalyzes the scrambling of phospholipids across the outer mitochondrial membrane; the mechanism is unrelated to channel activity and is capable of translocating both anionic and zwitterionic phospholipids.

 1 Publication

total 19 transmembrane segments

	TYPE	ID	POSITION(S)	DESCRIPTION	
1	+ Transmembrane		26-35	Beta stranded 2 Publications	Tools Add
2	+ Transmembrane		39-47	Beta stranded 2 Publications	Tools Add
3	+ Transmembrane		54-64	Beta stranded 2 Publications	Tools Add
4	+ Transmembrane		69-76	Beta stranded 2 Publications	Tools Add
5	Transmembrane		80-89	Beta stranded 2 Publications	Tools Add
6	+ Transmembrane		95-104	Beta stranded 2 Publications	Tools Add
7	Transmembrane		111-120	Beta stranded 2 Publications	Tools Add
8	+ Transmembrane		123-130	Beta stranded 2 Publications	Tools Add
9	Transmembrane		137-145	Beta stranded 2 Publications	Tools Add
10	+ Transmembrane		150-158	Beta stranded 2 Publications	Tools Add
11	Transmembrane		163-175	Beta stranded 2 Publications	Tools Add
12	+ Transmembrane		178-185	Beta stranded 2 Publications	Tools Add
13	Transmembrane		189-198	Beta stranded 2 Publications	Tools Add
14	+ Transmembrane		202-211	Beta stranded 2 Publications	Tools Add
15	Transmembrane		218-227	Beta stranded 2 Publications	Tools Add
16	+ Transmembrane		231-238	Beta stranded 2 Publications	Tools Add
17	Transmembrane		242-251	Beta stranded 2 Publications	Tools Add
18	+ Transmembrane		254-263	Beta stranded 2 Publications	Tools Add
19	Transmembrane		273-282	Beta stranded 2 Publications	Tools Add

P21796 is human VDAC1 (voltage-dependent anion-selective channel 1), and because it's a mitochondrial outer membrane beta-barrel, the 19 segments correspond to the antiparallel beta-strands that roll up to form the pore rather than typical alpha-helical transmembrane passes

Q.2

Advanced Searchⁱ

Searching in
UniRef

Sequence identity

Sequence identity
50%
Remove

Cluster name

AND
Cluster name
transcription factors
Remove

Add Field
Cancel
Search

Type * in the search box to search for all values for the selected field.

searched this:

UniRef (identity:0.5) AND (name:"transcripti" **Advanced** List Search    Help

UniRef 148 results

 Tools  Download (148)  Add View: Cards ☐ Table ☒  Customize columns

Cluster ID	Cluster name	Types	Size	Organisms
☐ UniRef50_A0A9Q0TUE3	Cluster: HOMEBOX PROTEIN TRANSCRIPTION FACTORS		2 members	Salix purpurea (Purple osier willow) Populus alba (White poplar)

148 total clusters

downloaded as excel and in the sheet

after doing text to column:

139	UniRef50	Cluster: N	UniProtKB	2	Chaenoc	89	0.5
140	UniRef50	Cluster: N	UniProtKB	1	Dallia pec	287	0.5
141	UniRef50	Cluster: N	UniProtKB	19	Anophele	387	0.5
142	UniRef50	Cluster: S	UniProtKB	1	Culex quir	311	0.5
143	UniRef50	Cluster: S	UniProtKB	2	Culex quir	145	0.5
144	UniRef50	Cluster: S	UniProtKB	1	Culex quir	148	0.5
145	UniRef50	Cluster: N	UniProtKB	6	Harpegna	199	0.5
146	UniRef50	Cluster: N	UniProtKB	2	Daphnia p	361	0.5
147	UniRef50	Cluster: G	UniProtKB	12	Saccharor	629	0.5
148	UniRef50	Cluster: G	UniProtKB	2	Candida g	622	0.5
149	UniRef50	Cluster: P	UniProtKB	1	Corethrel	244	0.5
150				317			
151							
152							
153							

sum is 317, so 317 is the count of sequences

 clustering compresses redundant entries so each cluster has at least one representative sequence, and summing up the cluster member counts in Excel gives back the total sequence space of 317 hits

Q.3

UniProt BLAST Align Peptide search ID mapping SPARQL UniRef ▼ homo sapiens Advanced | List Search

Taxonomy
Filter by taxonomy

UniRef 362,243 results

Tools ▼ Download (362k) Add View: Cards Table Customize columns Share ▼

Cluster ID	Cluster name	Types	Size	Organisms	Length	Identity
☐ UniRef100_A0A0K6G2Q5	Cluster: Dystrophin [Homo sapiens]		1 member	Rhizoctonia solani	969	UniRef100
☐ UniRef100_A0A8J2E3B5	Cluster: Mitochondrial (Homo sapiens)		1 member	Cotesia congregata (Parasitoid wasp)	135	UniRef100
☐ UniRef100_A0A8J2HG11	Cluster: Cytosolic (Homo sapiens)		1 member	Cotesia congregata (Parasitoid wasp)	394	UniRef100
☐ UniRef100_A0A8J2HIF4	Cluster:		1	Cotesia	595	UniRef100

answer:

- ◆ 100% (217,087)
- ◆ 90% (96,154)
- ◆ 50% (49,002)

in advanced search we will get a different result (slightly different):

UniRef 48,766 results

Tools ▼ Download (49k) Add View: Cards Table Customize columns Share ▼

Cluster ID	Cluster name	Types	Size	Organisms
☐ UniRef50_A0A023IPH8	Cluster: Integrin a4 subunit (Fragment)		6 members	Homo sapiens (Human) Aotus azarae (Azara's night monkey) Callithrix kuhlii (Wied's black-tufted-ear marmoset) Mico argentatus (Silvery marmoset) Callithrix penicillata (Black-tufted-ear marmoset) More organisms
☐ UniRef50_A0A023IQH3	Cluster: Integrin b7 subunit (Fragment)		1 member	Homo sapiens (Human)

at 50 percent sequence:

total number of clusters = 48,766

UniRef 95,627 results

Tools ▼ Download (96k) Add View: Cards Table Customize columns Share ▼

Cluster ID	Cluster name	Types	Size	Organisms	Length	Identity
☐ UniRef90_A0A023PPR2	Cluster: RUNX1/RUNX1T1 fusion protein		1 member	Homo sapiens (Human)	47	UniRef90
☐ UniRef90_A0A023T3F1	Cluster: PVT1/AKT3 fusion protein isoform 1 (Fragment)		1 member	Homo sapiens (Human)	59	UniRef90
☐ UniRef90_A0A023T4H3	Cluster: PVT1/EYA1 fusion protein		2 members	Homo sapiens (Human)	90	UniRef90

UniRef 215,857 results

Cluster ID	Cluster name	Types	Size	Organisms	Length	Identity
UniRef100_A0A023Q7P2	Cluster: Cytochrome c oxidase subunit 1		1 member	Homo sapiens (Human)	513	UniRef100
UniRef100_A0A023QEVO	Cluster: ATP synthase subunit a		1 member	Homo sapiens (Human)	226	UniRef100
UniRef100_A0A024A2P5	Cluster: SEC31A/ALK		1	Homo sapiens	66	UniRef100

as identity threshold drops from 100% down to 50%, sequence space collapses heavily because more divergent homologs get merged into shared clusters, and the slight variance in advanced search comes down to dynamic clustering updates vs static precomputed releases (UniRef50)

Q.4

Advanced Searchⁱ

Searching in
UniProtKB

Organism [OS]
10090 Remove

Reviewed
Yes Remove

AND Reviewed

Cancel Search

Type * in the search box to search for all values for the selected field.

this gives 17,267 results

UniProtKB 17,267 results

restrict search to proteome UP000000589

Entry	Entry Name	Protein Names	Gene Names	Organism
A0A5F8MPU3	CTSRT_MOUSE	Cation channel sperm-associated targeting subunit tau[...]	Catspert, Als2cr11, C2cd6	Mus musculus (Mouse)

Reviewed (Swiss-Prot) (17,267)

Popular organisms
Mouse (17,267)

Taxonomy
10090

Advanced Search ×

Searching in
UniProtKB

Organism [OS]
 × Remove

Reviewed
 AND Remove

3D Structure
 AND Remove

[Add Field](#) Cancel Search

*Type * in the search box to search for all values for the selected field.*

this gives 2,640 structures that are associated with 3d pdb

UniProt Tools UniProtKB (organism_id:10090) AND (reviewed:true) AND (struct Advanced List Search 📖 🔖 ✉ Help)

Status
 Reviewed (Swiss-Prot)
 (2,640)

Popular organisms
 Mouse (2,640)

Taxonomy
 10090 ×
[Filter by taxonomy](#)

Group by
 Taxonomy
 Keywords

UniProtKB 2,640 results

[Tools](#) [Download \(3k\)](#) [Customize columns](#) [Share](#)

Entry	Entry Name	Protein Names	Gene Names	Organism
☐ A2AIP0	CM12B_MOUSE	Ciliary microtubule inner protein 2B	Cimip2b, Fam166b	Mus musculus (Mouse)
☐ A2CG49	KALRN_MOUSE	Kalirin[...]	Kalrn	Mus musculus (Mouse)
☐ B2RXS4	PLXB2_MOUSE	Plexin-B2	Plxnb2	Mus musculus (Mouse)

out of the 17,267 entries found, only ~15% (2,640) have experimentally solved 3D structures deposited in the PDB, showing the typical gap between known sequence records and solved atomic coordinates

Q.5

UniProtKB 2,640 results
or restrict search to proteome UP000000589

Tools
Download (10)
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Share
10 rows selected out of 25

Entry	Entry Name	Protein Names	Gene Names
☒ A2AIP0	CMI2B_MOUSE	Ciliary microtubule inner protein 2B	Climip2b, Fam166b
☒ A2CG49	KALRN_MOUSE	Kalirin[...]	Kalrn
☒ B2RXS4	PLXB2_MOUSE	Plexin-B2	Plxbn2
☒ B5KM66	SYCE3_MOUSE	Synaptonemal complex central element protein 3[...]	Syce3, Tseg2
☒ E9Q238	KANK1_MOUSE	KN motif and ankyrin repeat domains 1[...]	Kank1
☒ E9Q7R9	CFA43_MOUSE	Cilia- and flagella-associated protein 43	Cfap43
☒ E9Q9F6	CTSRD_MOUSE	Cation channel sperm-associated auxiliary subunit delta[...]	Catsperd, Tmem146
☒ E9Q9W7	PDZD7_MOUSE	PDZ domain-containing protein 7	Pdzd7
☒ J3QMY9	TO6BL_MOUSE	Type 2 DNA topoisomerase 6 subunit B-like[...]	Top6bl, Gm960
☒ O08580	ERR1_MOUSE	Steroid hormone receptor ERR1[...]	Esrra, Err1, Estrra, Nr3b1
☐ O08582	GTPB1_MOUSE	GTP-binding protein 1[...]	Gtpbp1, Gtpbp
☐ O08762	NETR_MOUSE	Neurotrypsin[...]	Prss12, Bssp3
☐ O08807	PRDX4_MOUSE	Peroxiredoxin-4[...]	Prdx4
☐ O08908	P85B_MOUSE	Phosphatidylinositol 3-kinase regulatory subunit beta[...]	Plk3r2

Retrieve/ID mapping¹

Enter one or more IDs (100,000 max). You may also [load from a text file](#). Separate IDs by whitespace (space, tab, newline) or commas. This service can also be used [programmatically](#). Alternatively, the [underlying data](#) can be downloaded.

A2AIP0

A2CG49

B2RXS4

B5KM66

E9Q238

E9Q7R9

E9Q9F6

E9Q9W7

J3QMY9

O08580

Tool results

Your tool analysis results from the last 7 days are listed below. If you have tools jobs running, you can navigate away to other pages and you will be notified once the job is completed.

Job type	Name	Created	Status
ID MAPPING	A2AIP0 +9 UniProtKB_AC-ID -- STR	2026-08-25 18:14	Completed (9 hits)
9JKregfzC		Source database: UniProtKB AC/ID Target database: STRING	

Tools
Download (9)
View:
Cards
Table
Resubmit

9 IDs mapped to 9 results
1 ID not mapped:
A2AIP0

From	To
A2CG49	10090.ENSMUSP000000110611
B2RXS4	10090.ENSMUSP000000104955
B5KM66	10090.ENSMUSP000000131766
E9Q238	10090.ENSMUSP000000042177
E9Q7R9	10090.ENSMUSP000000125007
E9Q9F6	10090.ENSMUSP000000108603
E9Q9W7	10090.ENSMUSP000000133273
J3QMY9	10090.ENSMUSP000000136515
O08580	10090.ENSMUSP000000025906

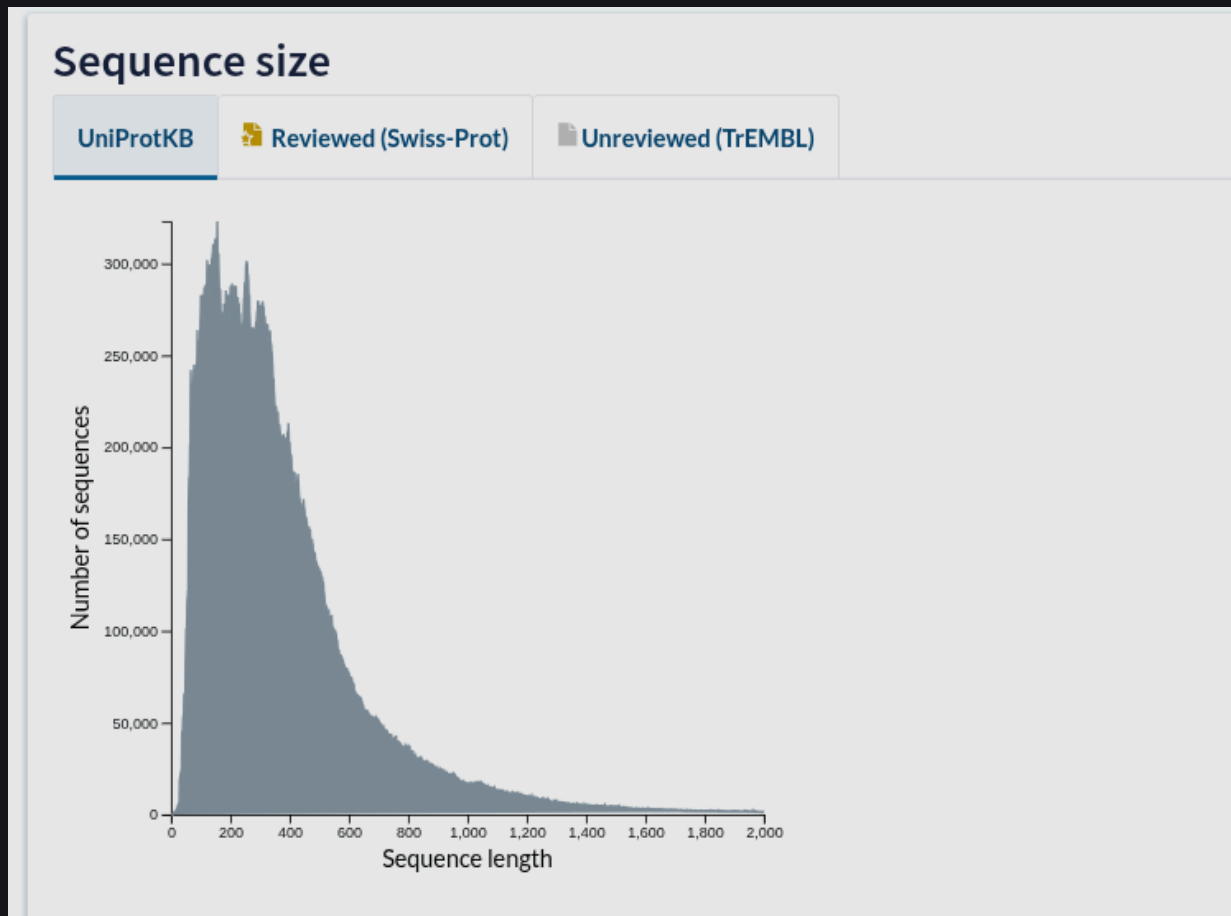
total 9 mapped, 1 didnt map

the 1 unmapped identifier usually happens due to obsolete secondary accession IDs, withdrawn entries,

or non-standard syntax that fails the UniProt ID mapping cross-reference table

Q.6

a.



The distribution is strongly right-skewed with a peak around 150 to 350 amino acids,

showing that most natural proteins are relatively compact and composed of just 1 or 2 structural domains (since a stable single folding unit is usually ~100-200 residues). There are very few tiny sequences under 50 residues because they struggle to form stable structures,

while giant proteins over 1000 residues make up the long thin tail because they have high costs of synthesising and greater risks of translational errors or misfolding.

from an evolutionary thermodynamics standpoint,
single-domain modules (~150-300 AA) minimize the

entropy cost of folding and reduce chaperone dependence during ribosomal translation

b.

UniProtKB 10,080 results

Entry	Entry Name	Protein Names	Gene Names	Organism	Length
☐ P0DPR3	TRDD1_HUMAN	T cell receptor delta diversity 1	TRDD1	Homo sapiens (Human)	2 AA
☐ P83570	GWA_SEPOF	Neuropeptide GWa		Sepia officinalis (Common cuttlefish)	2 AA
☐ P62968	TRH_PIG	Thyrotropin-releasing hormone[...]	TRH	Sus scrofa (Pig)	3 AA
☐ P62969	TRH_SHEEP	Thyrotropin-releasing hormone[...]	TRH	Ovis aries (Sheep)	3 AA
☐ P62970	TRH_BOMOR	Thyrotropin-releasing hormone[...]		Bombina orientalis (Oriental fire-bellied toad)	3 AA
☐ P62971	TRH_NOTVI	Thyrotropin-releasing hormone[...]		Notophthalmus viridescens (Eastern newt) (Triturus viridescens)	3 AA
☐ P84761	ACI_MACGN	Angiotensin-1-converting enzyme inhibitory peptide		Macrocybe gigantea (Giant mushroom) (Tricholoma giganteum)	3 AA
☐ P01858	TUFT_HUMAN	Phagocytosis-stimulating peptide[...]		Homo sapiens (Human)	4 AA
☐ P0DPI4	TDB01_HUMAN	T cell receptor beta diversity 1	TRBD1	Homo sapiens (Human)	4 AA
☐ P0DRH1	CRY26_TITOB	Cryptide Pep-26		Tityus obscurus (Amazonian scorpion) (Tityus cambridgei)	4 AA
☐ P0DRH2	CRY27_TITOB	Cryptide Pep-27		Tityus obscurus (Amazonian scorpion) (Tityus cambridgei)	4 AA

shortest is 2 AA

UniProtKB 1,539 results

Entry	Entry Name	Protein Names	Gene Names	Organism	Length
☐ A2ASS6	TITIN_MOUSE	Titin[...]	Ttn	Mus musculus (Mouse)	35,213 AA
☐ Q8WZ42	TITIN_HUMAN	Titin[...]	TTN	Homo sapiens (Human)	34,350 AA
☐ G4SLH0	TTN1_CAEEL	Titin homolog[...]	ttn-1, W06H8.8	Caenorhabditis elegans	18,562 AA
☐ Q9I7U4	TITIN_DROME	Titin[...]	sls, titin, CG1915	Drosophila melanogaster (Fruit fly)	18,141 AA
☐ AOA056XHH0	FRBI_DOTX1	FR901469 synthetase[...]	frbi, ANO11243_029940	Dothideomycetidae sp. (strain 11243) (Fungal sp. (strain No.11243))	15,639 AA
☐ Q09164	SIMA_TOLIN	Cyclosporin synthetase simA[...]	simA	Tolypocladium inflatum (Cyclosporin fungus) (Tolypocladium niveum)	15,281

largest is 35,213 AA

c.

acid	code			acid	UniProtKB entry
Leu	L	5,663,435,734	9.72%	149,601,225	37.80
Ala	A	5,098,843,511	8.75%	149,350,430	34.04
Ser	S	4,251,193,974	7.30%	149,426,993	28.38
Gly	G	4,118,666,200	7.07%	149,194,125	27.49
Val	V	3,947,571,082	6.78%	149,359,393	26.35
Glu	E	3,692,247,533	6.34%	148,679,847	24.65
Arg	R	3,437,397,109	5.90%	148,934,474	22.95
Thr	T	3,254,569,854	5.59%	149,134,222	21.72
Asp	D	3,190,792,267	5.48%	148,525,785	21.30
Pro	P	3,064,821,863	5.26%	148,220,928	20.46
Ile	I	2,995,660,501	5.14%	148,629,693	20.00
Lys	K	2,919,516,458	5.01%	146,418,483	19.49
Gln	Q	2,301,555,231	3.95%	147,505,443	15.36
Asn	N	2,205,644,882	3.79%	146,227,215	14.72
Phe	F	2,202,538,832	3.78%	147,321,587	14.70
Tyr	Y	1,627,739,696	2.79%	144,745,128	10.87
His	H	1,357,511,157	2.33%	143,216,000	9.06
Met	M	1,325,111,364	2.27%	149,100,901	8.85
Cys	C	848,209,881	1.46%	128,814,848	5.66
Trp	W	748,072,694	1.28%	131,978,372	4.99
Xaa	X	7,147,980	0.01%	780,155	0.05
Sec	U	8,965	<0.01%	7,928	<0.01
Asx	B	5,905	<0.01%	5,082	<0.01
Glx	Z	2,575	<0.01%	2,237	<0.01
Pyl	O	153	<0.01%	153	<0.01

